

5.3.2 Integration Worksheet

1. Evaluate $\int (6t^2 + 5e^t + 2) dt$

$$\int (6t^2 + 5e^t + 2) dt = 2t^3 + 5e^t + 2t + C$$

2. Evaluate $\int [10x^4 - 2\sec^2(x)] dx$

$$\int [10x^4 - 2\sec^2(x)] dx = \frac{10}{5}x^5 - 2\tan(x) + C = 2x^5 - 2\tan(x) + C$$

3. Evaluate $\int_{1/2}^{\sqrt{2}/2} -(1-x^2)^{-1/2} [\sqrt{1-x^2} + 1] dx$

$$\text{We note that } \int_{1/2}^{\sqrt{2}/2} -(1-x^2)^{-1/2} [\sqrt{1-x^2} + 1] dx = \int_{1/2}^{\sqrt{2}/2} -\frac{1}{\sqrt{1-x^2}} [\sqrt{1-x^2} + 1] dx = \int_{1/2}^{\sqrt{2}/2} \left[-1 + \frac{-1}{\sqrt{1-x^2}} \right] dx$$

Since $f(x) = -x - \frac{1}{\sqrt{1-x^2}}$ is continuous on $\left[\frac{1}{2}, \frac{\sqrt{2}}{2}\right]$ we know by the FTCP2 that:

$$\begin{aligned} \int_{1/2}^{\sqrt{2}/2} \left[-1 + \frac{-1}{\sqrt{1-x^2}} \right] dx &= \left[-x + \cos^{-1}(x) \right]_{1/2}^{\sqrt{2}/2} = \left[-\frac{\sqrt{2}}{2} + \cos^{-1}\left(\frac{\sqrt{2}}{2}\right) \right] - \left[-\frac{1}{2} + \cos^{-1}\left(\frac{1}{2}\right) \right] \\ &= -\frac{\sqrt{2}}{2} + \frac{\pi}{4} + \frac{1}{2} - \frac{\pi}{3} \\ &= \frac{1-\sqrt{2}}{2} - \frac{\pi}{12} \end{aligned}$$

4. Evaluate $\int (3x - 10\sqrt{x})^2 dx$.

$$\int (3x - 10\sqrt{x})^2 dx = \int (9x^2 - 60x\sqrt{x} + 100x) dx = \int (9x^2 - 60x^{3/2} + 100x) dx = 3x^3 - 24x^{5/2} + 50x^2 + C$$

5. Evaluate $\int_{-1}^1 \frac{x^4 + 3}{x^2 + 1} dx$. (Hint: Try to use a fancy 0!)

Since $f(x) = \frac{x^4 + 3}{x^2 + 1}$ is continuous on $[-1, 1]$ we know that we can use the FTC2. We note that:

$$\begin{aligned} \int_{-1}^1 \frac{x^4 + 3}{x^2 + 1} dx &= \int_{-1}^1 \frac{x^4 - 1 + 4}{x^2 + 1} dx = \int_{-1}^1 \left[\frac{x^4 - 1}{x^2 + 1} + \frac{4}{x^2 + 1} \right] dx = \int_{-1}^1 \left[\frac{(x^2 + 1)(x^2 - 1)}{x^2 + 1} + \frac{4}{x^2 + 1} \right] dx = \int_{-1}^1 \left[x^2 - 1 + \frac{4}{x^2 + 1} \right] dx \\ &= \left[\frac{1}{3}x^3 - x + 4 \tan^{-1}(x) \right]_{-1}^1 = \left[\frac{1}{3}(1)^3 - (1) + 4 \tan^{-1}(1) \right] - \left[\frac{1}{3}(-1)^3 - (-1) + 4 \tan^{-1}(-1) \right] \\ &= \left[\frac{1}{3} - 1 + 4 \left(\frac{\pi}{4} \right) \right] - \left[-\frac{1}{3} + 1 + 4 \left(-\frac{\pi}{4} \right) \right] \\ &= \left[-\frac{2}{3} + \pi \right] - \left[\frac{2}{3} - \pi \right] \\ &= -\frac{4}{3} + 2\pi \end{aligned}$$

6. Evaluate $\int \left(\frac{6x - x^3}{2x^2} \right) dx$.

$$\int \left(\frac{6x - x^3}{2x^2} \right) dx = \int \left(\frac{6x}{2x^2} - \frac{x^3}{2x^2} \right) dx = \int \left(\frac{3}{x} - \frac{1}{2}x \right) dx = 3 \ln|x| - \frac{1}{4}x^2 + C$$

7. Evaluate $\int_0^{2\pi} \tan(x) dx$.

We note that $f(x) = \tan(x)$ is NOT continuous on $[0, 2\pi]$, specifically because it is undefined at

$x = \frac{\pi}{2}$ & $\frac{3\pi}{2}$. Therefore, the FTC2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

8. Evaluate $\int \frac{4\sin^5(x) - 3[1 - \cos^2(x)]}{\sin^4(x)} dx$.

$$\begin{aligned} \int \frac{4\sin^5(x) - 3[1 - \cos^2(x)]}{\sin^4(x)} dx &= \int \frac{4\sin^5(x) - 3\sin^2(x)}{\sin^4(x)} dx = \int \left[\frac{4\sin^5(x)}{\sin^4(x)} - \frac{3\sin^2(x)}{\sin^4(x)} \right] dx \\ &= \int \left[4\sin(x) - \frac{3}{\sin^2(x)} \right] dx = \int [4\sin(x) - 3\csc^2(x)] dx \\ &= -4\cos(x) + 3\cot(x) + C \end{aligned}$$

9. Evaluate $\int_{\pi/6}^{\pi/3} [\csc(x) - \csc(x)\cos^2(x)] dx$.

Since $f(x) = \csc(x) - \csc(x)\cos^2(x)$ is continuous on $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ we know that we can use the FTC2.

We note that:

$$\begin{aligned} \int_{\pi/6}^{\pi/3} [\csc(x) - \csc(x)\cos^2(x)] dx &= \int_{\pi/6}^{\pi/3} \csc(x)[1 - \cos^2(x)] dx = \int_{\pi/6}^{\pi/3} \csc(x)\sin^2(x) dx = \int_{\pi/6}^{\pi/3} \sin(x) dx \\ &= -\cos(x) \Big|_{\pi/6}^{\pi/3} = -\cos\left(\frac{\pi}{3}\right) + \cos\left(\frac{\pi}{6}\right) \\ &= -\frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}-1}{2} \end{aligned}$$

10. Evaluate $\int \frac{\cos(2\theta) + \sin^2(\theta)}{\cos(\theta)} d\theta$.

$$\int \frac{\cos(2\theta) + \sin^2(\theta)}{\cos(\theta)} d\theta = \int \frac{\cos^2(\theta) - \sin^2(\theta) + \sin^2(\theta)}{\cos(\theta)} d\theta = \int \frac{\cos^2(\theta)}{\cos(\theta)} d\theta = \int \cos(\theta) d\theta = \sin(\theta) + C$$

11. Determine the average value of the function $f(x) = \frac{8}{x^2}$ on the interval $[1, 5]$.

We know that $f_{avg} = \frac{1}{5-1} \int_1^5 \frac{8}{x^2} dx$. Since $f(x) = \frac{8}{x^2}$ is continuous on $[1, 5]$ we know by the FTCP2 that:

$$f_{avg} = \frac{1}{5-1} \int_1^5 \frac{8}{x^2} dx = \frac{1}{4} \int_1^5 8x^{-2} dx = \frac{1}{4} \left(-8x^{-1} \right) \Big|_1^5 = -2(5)^{-1} + 2(1)^{-1} = -\frac{2}{5} + \frac{2}{1} = \frac{8}{5}$$

Therefore $f_{avg} = \frac{8}{5} = 1.6$.

12. Evaluate $\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$.

We note that $f(x) = \frac{1}{\sqrt{1-x^2}}$ is NOT continuous on $[0, 1]$, specifically because it is undefined at $x = 1$.

Therefore, the FTCP2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.